

Accuracy Report

Overview:

This presents a detailed evaluation of the Intelligent RAG Q&A System, focusing on accuracy, reliability, and performance. The assessment measured both retrieval and response accuracy using a set of structured queries targeting the private knowledge base. The system was tested for its ability to deliver contextually accurate, well-cited, and timely responses based on the internal data source.

Testing Methodology:

A total of **30 curated questions** were used to evaluate knowledge base accuracy. The test set included a mix of easy, medium, and hard questions spanning multiple categories such as model architecture, attention mechanisms, and security. Each query was assessed based on keyword coverage, factual accuracy, confidence score, and presence of citations. Response time and system stability were also recorded. Web fallback tests were skipped for this phase.

Definition of Accuracy:

For this hybrid RAG-based system, **accuracy** is defined as the ratio of successful responses that meet the following criteria:

- Correct retrieval of semantically relevant information from the private knowledge base.
- Keyword coverage ≥ 0.6 to ensure contextual completeness.
- Confidence score ≥ 0.9 for model assurance.
- Correct citation presence (inline or end-reference).
- Factual consistency with the original source material.

Results Summary:

- **Total Questions:** 30
- **Successful Responses:** 24
- **Failed Responses:** 6
- **Overall Success Rate:** 80.0%
- **Average Response Time:** 10.98 seconds
- **Average Keyword Coverage:** 64.3%
- **Average Confidence:** 0.918
- **Responses with Sources:** 100%
- **Responses with Inline Citations:** 63.3%
- **Grade:** B

Difficulty Breakdown:

- **Easy:** 3/3 successful (100%)
- **Medium:** 11/14 successful (78.6%)
- **Hard:** 10/13 successful (76.9%)

Category Breakdown:

- Architecture Components: 4/4 successful
- Attack Methods: 3/3 successful
- Architecture Basics: 1/1 successful
- Architecture Details: 2/2 successful
- Model Comparison: 0/1 failed
- Security Threats: 0/1 failed
- Architecture Features: 0/1 failed
- Classification: 0/1 failed
- All other categories achieved ≥ 1 successful response

Performance Metrics:

- **Average Latency:** 10.98 seconds
- **Minimum Latency:** 10.35 seconds
- **Maximum Latency:** 11.78 seconds
- **Backend Health:** Stable (0 errors)
- **Backend Response Time:** 0.010 seconds

Common Failure Causes:

- Low keyword coverage in complex, multi-layer reasoning questions.
- Missed contextual links for abstract queries (e.g., hardware classification and security implications).
- Minor degradation in attention-mechanism-related answers due to token misalignment.
- Some responses lacked inline citations despite correct source attribution.

Test Environment:

All tests were executed within the system's integrated development setup using the following configuration:

- **Vector Database:** Qdrant (local instance)
- **Embedding Model:** BAAI/bge-small-en-v1.5
- **Language Model (LLM):** Gpt-oss-120B
- **Frameworks:** LangChain and LangGraph for orchestration
- **Evaluation Pipeline:** Python-based automated testing suite for response scoring and logging

Conclusion:

The Intelligent RAG Q&A System achieved an **80% accuracy rate** across a diverse set of knowledge-based queries, indicating strong reliability in retrieval and synthesis. The model demonstrates consistent confidence and maintains factual correctness in most responses. Areas for improvement include enhanced keyword recall, latency reduction, and inline citation consistency. Overall, the system meets expected performance benchmarks.